Exercise 2

Name: Harini Nagarajan Roll Number:240701167

CLI, GUI, and VUI Implementation Using Python

AIM:

The aim is to develop Command Line Interface (CLI), Graphical User Interface (GUI), and Voice User Interface (VUI) using Python (with Tkinter for GUI and Speech Recognition for VUI) and Terminal.

PROCEDURE:

CLI (Command Line Interface)

```
import os

def find_file_auto():
    print("🔍 Universal File Finder")

    file_name = input("Enter file name (with extension): ").strip().lower()

    search_path = os.path.expanduser("~")

    print(f"\nSearching in: {search_path}")
print("Please wait...\n")

    matches = []

    for root, dirs, files in os.walk(search_path):
        for file in files:
            if file.lower() == file_name:  # case-insensitive match
                matches.append(os.path.join(root, file))

    if matches:
    print(f"✅ Found {len(matches)} matches:\n")
      for path in matches:
```

```
                print(path)
            else:
            print("❌ File not found.")
    find_file_auto()
```

Sample Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

' 'c:\Users\HARINI\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '62645' '--' 'c:\Users\HARINI\Desktop\ui python experiments\exp3.py'
🔍 Universal File Finder
Enter file name (with extension): java programs.pdf

Searching in: C:\Users\HARINI
Please wait...

✅ Found 1 matches:

C:\Users\HARINI\Documents\java programs.pdf

Python
Python
powersh...
powersh...
Python Deb...
```

<u>GUI (Graphical User Interface)</u>

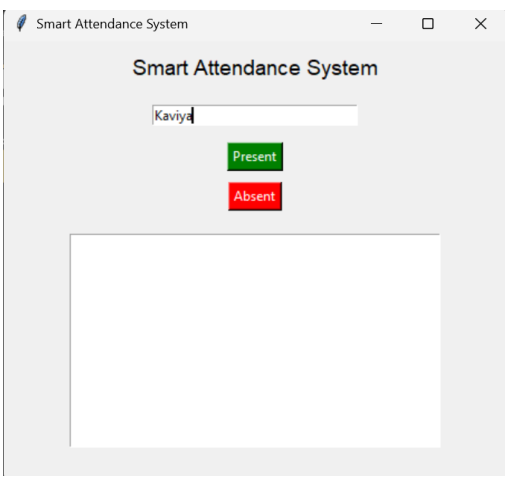
Tkinter to create a simple GUI for Smart Attendance System

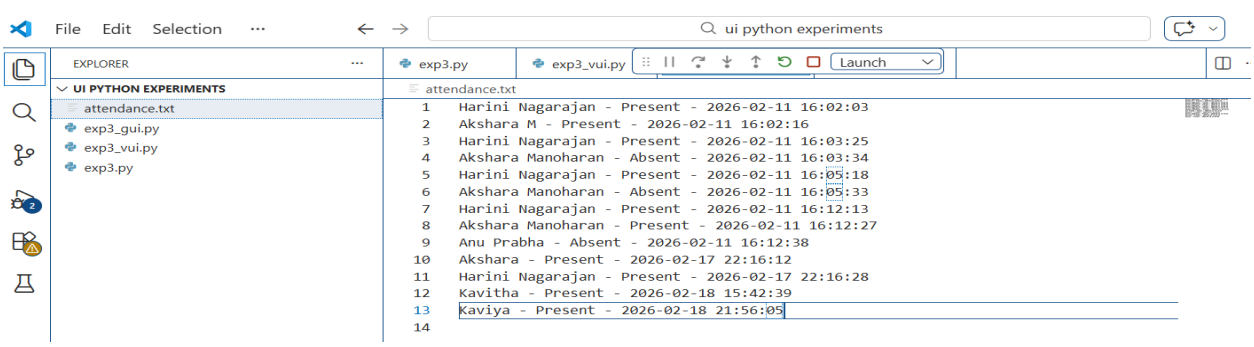
```
import tkinter as tk
from tkinter import messagebox
from datetime import datetime


def mark_present():
    name = entry.get()


    if name == "":
        messagebox.showwarning("Warning", "Enter student name")
    else:
        time = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
        record = name + " - Present - " + time
```

```
        listbox.insert(tk.END, record)
        file = open("attendance.txt", "a")
        file.write(record + "\n")
        file.close()
        entry.delete(0, tk.END)

def mark_absent():
    name = entry.get()

    if name == "":
        messagebox.showwarning("Warning", "Enter student name")
    else:
        time = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
        record = name + " - Absent - " + time
        listbox.insert(tk.END, record)
        file = open("attendance.txt", "a")
        file.write(record + "\n")
        file.close()
         entry.delete(0, tk.END)
root = tk.Tk()
root.title("Smart Attendance System")
root.geometry("450x400")
label = tk.Label(root, text="Smart Attendance System", font=("Arial", 14))
label.pack(pady=10)
```

```
entry = tk.Entry(root, width=30)

entry.pack(pady=10)

present_btn = tk.Button(root, text="Present", bg="green", fg="white",
command=mark_present)

present_btn.pack(pady=5)

absent_btn = tk.Button(root, text="Absent", bg="red", fg="white",
command=mark_absent)

absent_btn.pack(pady=5)

listbox = tk.Listbox(root, width=55, height=12)

listbox.pack(pady=15)

root.mainloop()
```

Sample Output:





VUI (Voice User Interface)

```
import os
import speech_recognition as sr
import pyttsx3

engine = pyttsx3.init()
engine.setProperty('rate', 180)  # faster speech

def speak(text):
    engine.say(text)
    engine.runAndWait()

def listen():
    r = sr.Recognizer()
    r.pause_threshold = 0.8  # faster response

    with sr.Microphone() as source:
        print("🎤 Speak file name...")
        speak("Say the file name")
        audio = r.listen(source, phrase_time_limit=3)

    try:
        text = r.recognize_google(audio)
        print("You said:", text)
```

```
        return text.lower()
    except:
        speak("Could not understand")
        return None

def find_file():
    file_name = listen()
    if not file_name:
        return

    file_name = file_name.replace(" dot ", ".").replace("dot", ".")

    # 🔥 FAST SEARCH PATHS (instead of whole system)
    search_paths = [
        os.path.join(os.path.expanduser("~"), "Desktop"),
        os.path.join(os.path.expanduser("~"), "Downloads"),
        os.path.join(os.path.expanduser("~"), "Documents"),
    ]

    print("🔍 Searching quickly...")
    speak("Searching")

    matches = []
```

```
    for path in search_paths:
        if os.path.exists(path):
            for root, dirs, files in os.walk(path):
                for file in files:
                    if file.lower() == file_name:
                        matches.append(os.path.join(root, file))

    if matches:
        print("✅ Found:")
        speak("File found")
        for m in matches:
            print(m)
    else:
        print("❌ File not found")
        speak("File not found")

find_file()
```

Sample Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\HARINI\Desktop\ui python experiments> ^C
PS C:\Users\HARINI\Desktop\ui python experiments>
PS C:\Users\HARINI\Desktop\ui python experiments>  c:; cd 'c:\Users\HARINI\Desktop\ui pyth
on experiments'; & 'c:\Users\HARINI\anaconda3\python.exe' 'c:\Users\HARINI\.vscode\extensi
ons\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '55292' '--' 'c:\
Users\HARINI\Desktop\ui python experiments\exp3_vui.py'
🎤 Speak file name...
You said: Java programs.pdf
🔍 Searching quickly...
✅ Found:
C:\Users\HARINI\Documents\java programs.pdf

Python
Python
powersh...
powersh...
Python Deb...
Python Deb...
Python Deb...
Python Deb...
Python Deb...
```